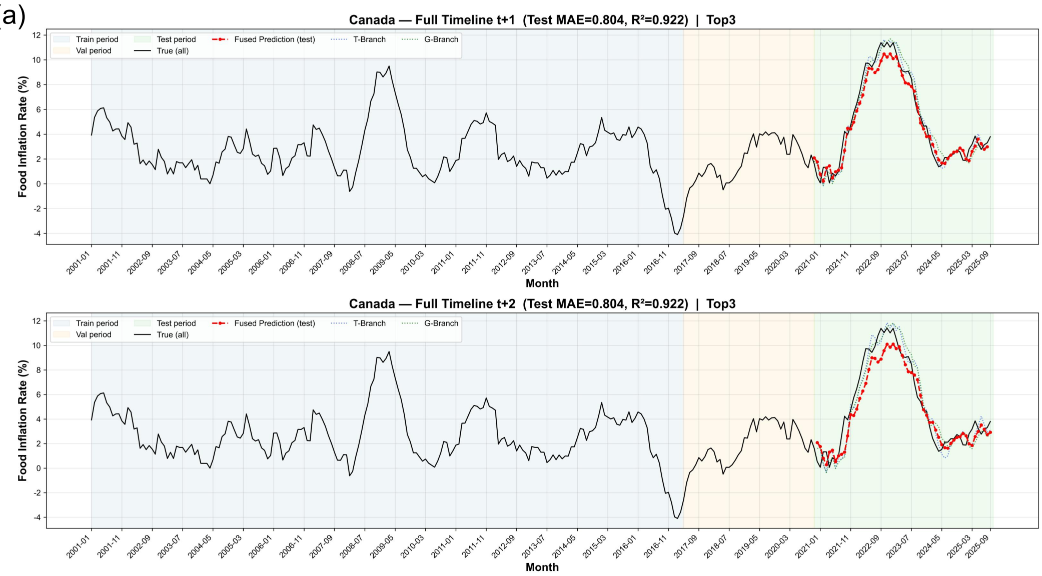
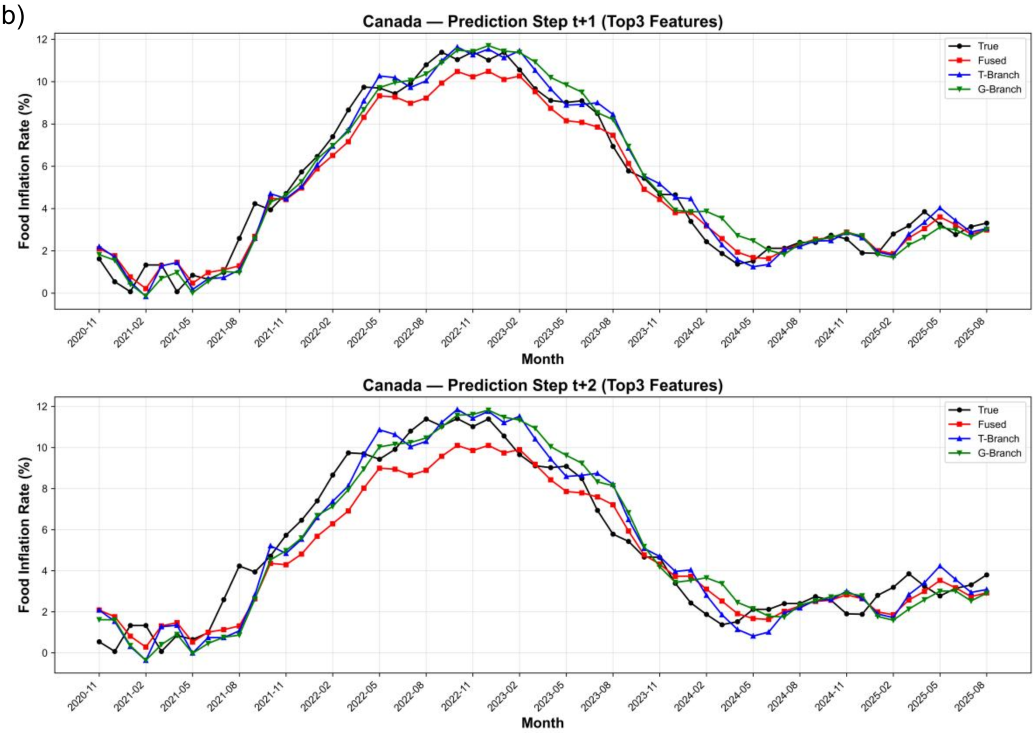
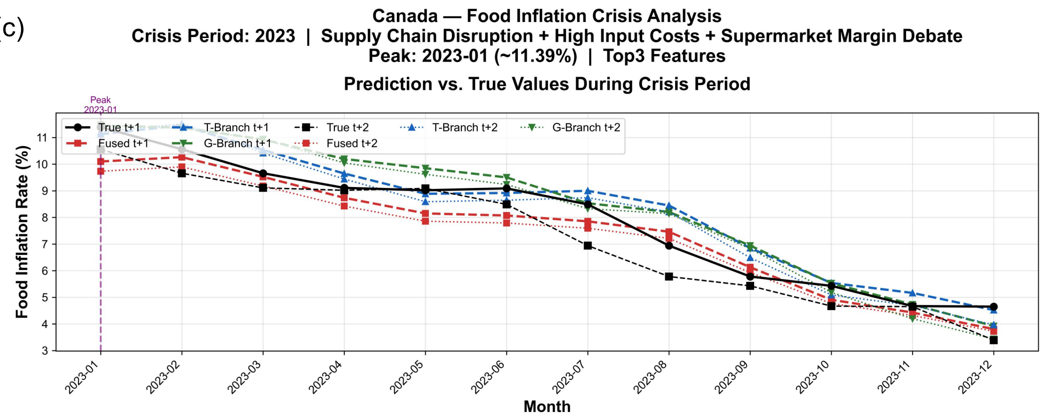
(a)

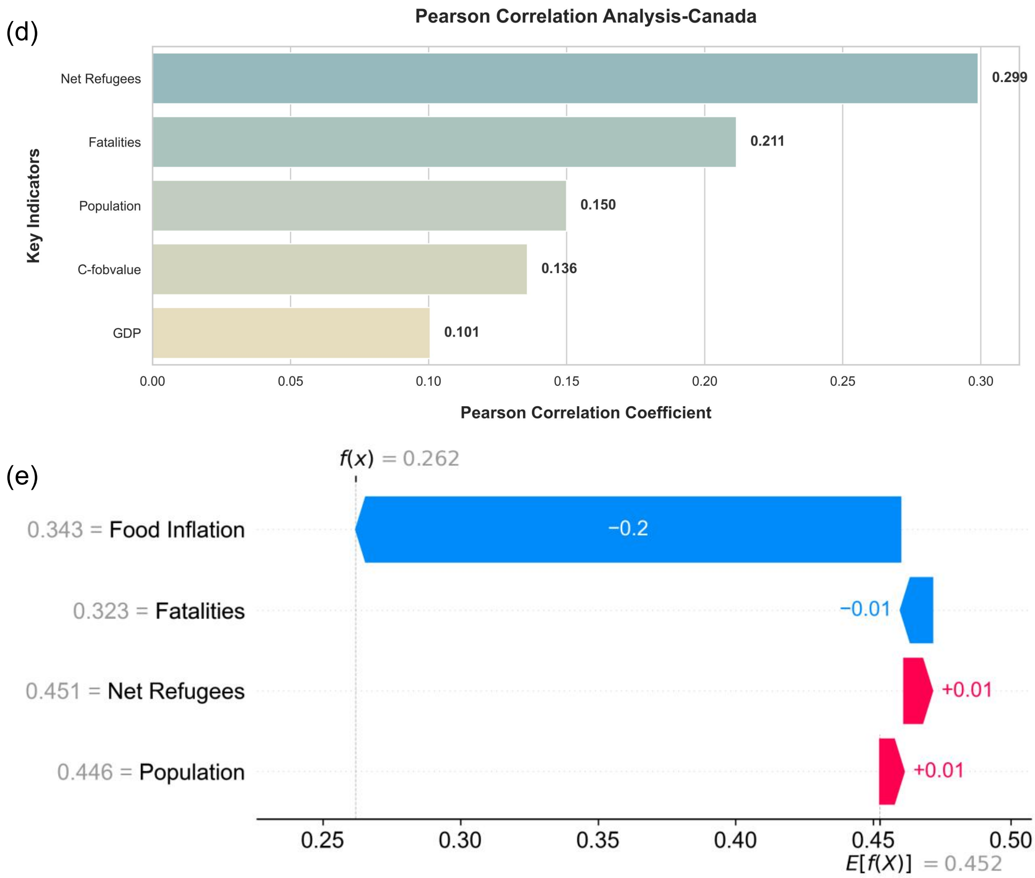

Canada — Full Timeline t+1 (Test MAE=0.804, R²=0.922) | Top3
Train period
Val period
Test period
True (all)
Fused Prediction (test)
T-Branch
G-Branch
Food Inflation Rate (%)
Month
Canada — Full Timeline t+2 (Test MAE=0.804, R²=0.922) | Top3
Train period
Val period
Test period
True (all)
Fused Prediction (test)
T-Branch
G-Branch
Food Inflation Rate (%)
Month

(b)


Canada — Prediction Step t+1 (Top3 Features)
True
Fused
T-Branch
G-Branch
Food Inflation Rate (%)
Month
Canada — Prediction Step t+2 (Top3 Features)
True
Fused
T-Branch
G-Branch
Food Inflation Rate (%)
Month

(c)


Canada — Food Inflation Crisis Analysis
Crisis Period: 2023 | Supply Chain Disruption + High Input Costs + Supermarket Margin Debate
Peak: 2023-01 (~11.39%) | Top3 Features
Prediction vs. True Values During Crisis Period
Peak
2023-01
True t+1
Fused t+1
T-Branch t+1
G-Branch t+1
True t+2
Fused t+2
T-Branch t+2
G-Branch t+2
Food Inflation Rate (%)
Month



Multidimensional Evaluation and Attribution Analysis of Food Inflation Forecast in Canada. (a) Overview of the total cycle prediction and fitting, (b) Multi-step Short-Term Forecast Comparison Chart, (c) Details of Prediction Performance in Extreme Crisis Periods, (d) Pearson correlation analysis of macroeconomic drivers, (e) SHAP-based explanation of feature contributions for a specific prediction instance.

As a representative high-latitude developed economy, food inflation in Canada is characterized by a pronounced seasonality and structural cost rigidity. Due to extended winters, a diminished self-sufficiency rate for fresh produce renders the price system highly sensitive to climatic shocks and cross-border logistics costs from southern North American production hubs, creating a distinctive winter vegetable-inflation cycle. Regarding economic transmission, mature commercial contracts often delay the immediate impact of global commodity shocks on retail prices, which propagate through extended processing, transportation, and packaging chains. Furthermore, the Canadian food retail market exhibits clear oligopolistic characteristics. The concentrated pricing power held by major supermarket chains enables the efficient pass-through of rising energy and labor costs to consumers, resulting in rapid upward price adjustments but significant downward stickiness. This interplay between retail profit expansion and supply chain transmission mechanisms is the fundamental driver of inflation in developed economies.